AI-Based Water Quality Monitoring Pool Management System

Project ID – R26-IT-143

Project Proposal Report

Nilogini Yogarasa

Water Quality Classification and prediction Module

B.Sc(Hons) Degree in Information Technology Specialized in

Information Technology

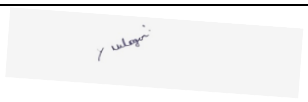
Department of Information Technology

Sri Lanka Institute of Information Technology
Srilanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Y.Nilogini	IT22247254	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Prof. Anuradha Karunasena	13/03/2026	

2. ABSTRACT

This research proposes the development of an AI-Based Water Quality Monitoring Pool Management System
designed to ensure safe swimming pool conditions through automated water quality analysis and monitoring. Traditional pool maintenance methods rely heavily on manual testing and fixed threshold monitoring, which can delay the detection of unsafe water conditions and increase health risks for pool users. To address these limitations, this study introduces a machine learning–based module capable of analyzing multiple water quality parameters in real time.

The proposed system utilizes IoT-based sensors to continuously collect water quality data, including pH**,** turbidity**,** total dissolved solids (TDS), temperature, and chlorine levels. These sensor readings are processed through machine learning algorithms to classify pool water conditions into three categories: Safe, Warning, and Critical. By analyzing the relationships between different water parameters, the system provides a more intelligent and data-driven approach to monitoring water quality compared to conventional threshold-based methods.

In addition to classification, the system performs water quality parameter importance analysis to determine the relative influence of each parameter in the decision-making process. This analysis helps identify the most significant factors affecting water safety, such as chlorine concentration or pH imbalance, allowing pool operators to focus on critical maintenance actions.

By integrating IoT-based monitoring with machine learning analytics, the proposed solution improves the accuracy and efficiency of pool water management. The system supports early detection of unsafe water conditions, enhances operational efficiency, and reduces reliance on manual monitoring processes. Ultimately, the research contributes to the development of smarter, safer, and more sustainable pool management systems suitable for public swimming pools, hotels, sports complexes, and recreational facilities.

Keywords: Artificial Intelligence, Water Quality Monitoring, Machine Learning, IoT Sensors, Pool Safety, Water Classification, Smart Pool Management.

TABLE OF CONTENTS

1. A DECLARATION, AND THE STATEMENT OF THE SUPERVISOR.....**Error! Bookmark not defined.**
2. ABSTRACT..... 3
TABLE OF CONTENTS..... 4
LIST OF FIGURES..... 6
LIST OF TABLES..... 6
1. INTRODUCTION..... 8
1.1 Background & Literature Survey..... 8
1.2 Research Gap..... 9
1.3 Research Problem..... 11
2. OBJECTIVE..... 12
2.1 Main Objective..... 12
2.2 Specific Objectives..... 12
2. Data Preprocessing and Feature Preparation..... 13
3. Machine Learning-Based Water Quality Classification..... 13
4. Water Quality Parameter Importance Analysis..... 13
5. Intelligent Monitoring and Decision Support..... 13
3. METHODOLOGY..... 13
Data Collection and Preprocessing..... 14
Water Quality Classification Using Machine Learning..... 14
Water Quality Parameter Importance Analysis..... 14
System Integration and Monitoring Interface..... 15
2.3 System Architecture..... 16
2.3.1 Software Solution..... 16
3.2 The Flow of the Project..... 16
3.2.1 Requirement Gathering and Analysis..... 16
Review of Literature and Existing Solutions..... 16
Stakeholder Feedback and System Requirements..... 16

Requirement Validation 17
3.2.2 Feasibility Study 17
Schedule Feasibility 17
Technical Feasibility 17
Operational Feasibility 17
3.2.3 Implementation 18
IoT Sensor Deployment and Data Collection 18
Data Preprocessing and Feature Extraction 18
Water Quality Classification Modeling 19
Parameter Importance Analysis 19
Monitoring Dashboard and System Integration 19
3.2.4 Testing 19
Unit Testing 19
Integration Testing 20
System Validation 20
Optimization and Refinement 20
3.3 Project Requirements 20
3.3.1 Functional Requirements 20
Real-Time Water Quality Monitoring 20
Water Quality Classification 20
Parameter Importance Analysis 21
Monitoring Dashboard 21
Data Storage and Reporting 21
System Integration 21
3.3.2 Non-Functional Requirements 21
Reliability 21
Performance 21
Usability 21
Scalability 22
Security 22

3.3.3 Software Requirements 22
Python / Pandas / NumPy / Scikit-learn 22
Machine Learning Frameworks 22
Data Visualization Tools 22
Database Management System 22
IoT Data Integration 23
Development Environment 23
Version Control System 23
3. DESCRIPTION OF PERSONAL AND FACILITIES 23
4. COMMERCIALIZATION 23
BUDGET AND BUDGET JUSTIFICATION 24
5. REFERENCE LIST 25
APPENDICES 26
Gantt chart 26
Work Breakdown Structure 27

LIST OF FIGURES

Figure1: Flow of Methodology 15
Figure 2: System Architecture Diagram 16
Figure 3: Gantt chart 28
Figure 4: Work Breakdown Structure 29

LIST OF TABLES

Table 1: Novelty of the Proposed System 09
Table 2: Budget 25

LIST OF ABBREVIATIONS

Abbreviation	**Meaning**
AI	Artificial Intelligence
IoT	Internet of Things
ML	Machine Learning
pH	Potential of Hydrogen
WQMS	Water Quality Monitoring System
RP	Research Project
NSF	Novel System Feature

1. INTRODUCTION

1.1 Background & Literature Survey

The rapid advancement of **Artificial Intelligence (AI)** and **Internet of Things (IoT)** technologies has significantly transformed the way environmental monitoring systems operate. These technologies enable continuous data collection, real-time analysis, and intelligent decision-making across many domains such as healthcare, agriculture, smart cities, and water management. In the context of swimming pool management, maintaining safe water quality is essential to ensure the health and safety of users. Poorly maintained pool water can lead to several health risks including skin infections, eye irritation, respiratory problems, and the spread of waterborne diseases.

Traditional swimming pool maintenance methods rely mainly on **manual testing and periodic chemical adjustments**. Pool operators typically measure parameters such as **pH level, chlorine concentration, turbidity, and temperature** using handheld test kits. Although these methods are widely used, they are often time-consuming, error-prone, and unable to provide continuous monitoring. As a result, unsafe water conditions may remain undetected for long periods, increasing the risk of contamination and poor water quality.

Recent developments in **IoT-based environmental monitoring systems** have introduced automated approaches for measuring water quality parameters using sensor technologies. Sensors can continuously monitor critical parameters such as **pH, turbidity, temperature, and total dissolved solids (TDS)** and transmit the collected data to centralized monitoring platforms. These systems provide real-time insights into water conditions and reduce the reliance on manual inspection.

In addition to IoT monitoring, **machine learning (ML) techniques** have shown strong potential in analyzing environmental data and identifying complex patterns that cannot be easily detected using traditional rule-based systems. Machine learning algorithms such as **Random Forest, Support Vector Machines (SVM), and Decision Trees** are widely used for classification and prediction tasks in environmental monitoring applications. By analyzing historical and real-time sensor data, these models can classify water conditions and detect abnormal patterns that may indicate contamination or unsafe water quality.

Several studies have demonstrated the effectiveness of machine learning models in water quality analysis and environmental monitoring. For example, research on water monitoring systems has shown that combining **sensor-based data collection with machine learning algorithms** significantly improves the accuracy of water quality classification and anomaly detection. These approaches highlight the importance of **data preprocessing, feature selection, and model evaluation** to ensure reliable predictions.

Despite these technological advancements, many existing swimming pool monitoring systems still rely on **simple threshold-based alerts** rather than intelligent data analysis. These systems can detect unsafe conditions only after the parameters exceed predefined limits, which limits their ability to provide early warnings or deeper insights into water quality factors.

Therefore, there is a need for a more **intelligent and data-driven pool monitoring solution** that integrates IoT sensor data with machine learning analysis. This research aims to address this gap by developing an **AI-based water quality classification module** that analyzes multiple sensor parameters and classifies pool water conditions into **Safe, Warning, and Critical categories**. Additionally, the system performs **parameter importance analysis** to identify which water quality factors most significantly influence water safety. By integrating sensor-based monitoring with machine learning techniques, the proposed system aims to enhance **pool safety, operational efficiency, and sustainable water management**.

1.2 Research Gap

Although recent advancements in Artificial Intelligence **(**AI**)** and Internet of Things **(**IoT**)** technologies have significantly improved environmental monitoring systems, there remains a noticeable gap in the development of intelligent and fully integrated swimming pool monitoring solutions. Most existing pool monitoring systems focus primarily on basic parameter measurement such as pH, chlorine levels, and temperature. While these systems provide useful information, they often rely on simple threshold based monitoring, which only triggers alerts when parameters exceed predefined limits.

Current commercial pool monitoring products typically use IoT sensors to measure water quality parameters and display the results through mobile applications. However, these systems rarely incorporate **machine learning techniques capable of analyzing complex relationships between multiple water quality variables**. As a result, they are limited in their ability to provide intelligent insights, predictive analysis, or deeper understanding of the factors influencing water quality conditions.

Furthermore, many existing systems operate as **standalone monitoring tools** that simply report sensor values without performing advanced data analysis. These systems lack the ability to classify water conditions automatically or identify the relative importance of different water parameters. Consequently, pool operators must still manually interpret sensor readings and determine appropriate maintenance actions, which can lead to delayed responses and inefficient pool management.

Several research studies have explored the application of machine learning techniques in environmental monitoring and water quality assessment. These studies demonstrate that **combining sensor-based data collection with machine learning algorithms** can significantly improve the accuracy of water quality classification and anomaly detection. However, many of these studies focus primarily on **natural water bodies such as rivers, lakes, and reservoirs**, rather than controlled environments like swimming pools.

Another limitation of existing solutions is the **lack of intelligent parameter analysis** that identifies which water quality variables have the greatest impact on overall pool safety. Understanding the relative importance of parameters such as chlorine levels, turbidity, pH, and temperature can help operators prioritize maintenance activities and optimize chemical usage.

Therefore, the research gap lies in the absence of an **AI-driven pool monitoring system that integrates real-time IoT sensor data with machine learning–based water quality classification and parameter importance analysis**. Such a system would not only monitor water quality continuously but also provide intelligent insights that support faster decision-making and improved pool safety management.

This research aims to address this gap by developing an **AI-based water quality classification module** capable of analyzing multiple sensor parameters and categorizing pool water conditions into **Safe, Warning, and Critical states**, while also identifying the most influential water quality factors affecting pool safety.

Feature	Existing Pool Monitoring Systems	**Basic ML Models**	**Research Prototypes**	**Our System**
Real-time water quality monitoring	Partial	Limited	Yes	Yes (IoT-based realtime monitoring)
Water quality classification	No	Yes(limited accuracy)	Partial	Yes (ML-based classification)
Multi-parameter sensor analysis	Limited	Yes	Yes	Yes (pH, turbidity, TDS, temperature, chlorine)
Parameter importance analysis	No	No	Rare	Yes (identifies key influencing parameters)
Intelligent decision support	No	No	Partial	Yes (data-driven insights for pool operators)
Integrated monitoring and analysis	No	Partial	Partial	Yes (fully integrated AI monitoring system)

Table 1: Novelty of the Proposed System

Table 1 presents a comparison between existing swimming pool monitoring approaches and the proposed AI-based intelligent pool monitoring system. Traditional pool monitoring systems mainly rely on sensorbased measurements and threshold-based alerts to monitor parameters such as pH, chlorine levels, and temperature. While these systems can provide basic real-time monitoring, they often lack advanced analytical capabilities and intelligent decision support.

Basic machine learning models have been applied in environmental monitoring applications to classify water quality conditions; however, many of these models operate independently from real-time

monitoring systems and often focus on limited parameters. Research prototypes demonstrate the potential of combining sensor data with machine learning techniques, but many lack complete system integration and practical deployment in real-world pool environments.

The proposed system improves upon these limitations by introducing an **integrated AI-based monitoring framework** that combines IoT-based real-time sensor data collection with machine learning–based water quality classification. In addition, the system performs **parameter importance analysis**, which identifies the most influential water quality parameters affecting pool safety. By integrating monitoring, classification, and intelligent analysis into a single system, the proposed solution provides a more effective and reliable approach to pool water quality management.

1.3 Research Problem

Maintaining safe water quality in swimming pools is essential to protect public health and ensure a safe recreational environment. Swimming pools in hotels, sports complexes, schools, and public facilities are used by a large number of people daily. As a result, water quality conditions can change rapidly due to factors such as swimmer activity, environmental contamination, and chemical imbalances. Poorly maintained pool water can lead to health issues including skin infections, eye irritation, respiratory problems, and the spread of waterborne diseases.

Traditional pool maintenance methods mainly rely on **manual testing and periodic monitoring of chemical parameters** such as pH level, chlorine concentration, turbidity, and temperature. Pool operators typically use handheld testing kits to measure these parameters at specific time intervals. However, these manual processes are often **time-consuming, inconsistent, and unable to provide continuous monitoring** of water conditions. As a result, unsafe water conditions may remain undetected for long periods.

Another significant challenge is the **lack of intelligent data analysis in existing pool monitoring systems**. Most commercial pool monitoring solutions rely on simple threshold-based alerts that notify operators only when sensor values exceed predefined limits. While this approach can detect extreme conditions, it does not analyze the relationships between multiple water quality parameters or provide deeper insights into the factors affecting pool safety.

Furthermore, many current systems operate as **basic monitoring tools rather than intelligent decisionsupport systems**. Pool operators often need to manually interpret sensor data and determine appropriate chemical adjustments or maintenance actions. This increases the risk of delayed responses, inefficient chemical usage, and improper pool maintenance.

Another limitation is the **absence of advanced analytical methods that identify the most influential water quality parameters** affecting pool safety. Understanding the relative importance of parameters such as chlorine levels, pH balance, turbidity, and temperature can help operators prioritize maintenance activities and optimize chemical treatment strategies.

Therefore, the key research problem is the **lack of an intelligent pool monitoring system that integrates real-time sensor data with machine learning-based water quality analysis and classification**. Without such a system, pool management remains largely reactive rather than proactive.

To address this issue, this research proposes the development of an **AI-based water quality classification module** that uses machine learning techniques to analyze multiple sensor parameters and classify pool water conditions into **Safe, Warning, and Critical states**. In addition, the system performs **parameter importance analysis** to identify the most significant factors influencing water quality. By providing continuous monitoring and intelligent analysis, the proposed system aims to improve pool safety, enhance operational efficiency, and support more effective pool management practices.

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to develop an **AI-based water quality classification module** for an intelligent swimming pool monitoring system. The proposed system will utilize IoT sensors to collect real-time water quality data such as **pH, turbidity, total dissolved solids (TDS), temperature, and chlorine levels**. Machine learning techniques will then be applied to analyze the collected sensor data and classify pool water conditions into **Safe, Warning, and Critical categories**.

By integrating **real-time sensor monitoring with machine learning-based data analysis**, the system aims to improve pool safety and provide a more intelligent approach to water quality management. The proposed solution seeks to reduce reliance on manual testing methods, enhance the efficiency of pool maintenance operations, and support proactive decision-making for pool operators.

2.2 Specific Objectives

To achieve the main objective of developing an intelligent water quality monitoring system, this research focuses on the following specific objectives:

1. Sensor-Based Water Quality Data Collection

The system will collect real-time water quality data using IoT sensors that measure parameters such as **pH, turbidity, temperature, total dissolved solids (TDS), and chlorine levels**. Continuous data collection ensures that the monitoring system has accurate and up-to-date information about pool conditions.

2. Data Preprocessing and Feature Preparation

The collected sensor data will be cleaned and processed to remove noise, handle missing values, and normalize parameter values. This step ensures that the dataset is suitable for training machine learning models and improves the reliability of classification results.

3. Machine Learning-Based Water Quality Classification

Machine learning algorithms such as **Random Forest, Decision Tree, or Support Vector Machines (SVM)** will be trained to classify pool water conditions into **Safe, Warning, and Critical categories** based on the collected sensor data. This classification helps pool operators quickly identify unsafe conditions.

4. Water Quality Parameter Importance Analysis

The system will analyze the relative importance of different water quality parameters to determine which variables most significantly influence the classification results. This analysis helps identify critical parameters such as chlorine concentration or pH imbalance that require immediate attention.

5. Intelligent Monitoring and Decision Support

The proposed system will provide insights and monitoring results that assist pool operators in making timely maintenance decisions. By presenting classified water quality status and parameter importance information, the system supports efficient pool management and improved safety.

3. METHODOLOGY

The methodology for developing the **AI-based water quality classification module** involves several stages designed to ensure accurate monitoring, reliable analysis, and effective decision support for swimming pool management. The system will collect real-time water quality data from IoT sensors and apply machine learning algorithms to analyze the collected data and classify water conditions.

The proposed approach begins with the **collection of water quality data** from sensors installed in the swimming pool environment. Key parameters monitored include **pH, turbidity, temperature, total dissolved solids (TDS), and chlorine levels**. In addition to real-time sensor readings, historical water quality data may also be used to train and validate machine learning models. Before analysis, the collected data will undergo preprocessing steps such as cleaning, formatting, and normalization to ensure consistency and suitability for machine learning applications.

Data Collection and Preprocessing

The first stage involves collecting continuous water quality data using **IoT-based sensors** placed in the swimming pool environment. These sensors measure important chemical and physical parameters including pH level, turbidity, chlorine concentration, temperature, and total dissolved solids.

The collected data will be transmitted to a central processing system where preprocessing operations will be performed. These preprocessing steps include:

- Handling missing or inconsistent values
- Removing noise from sensor readings
- Normalizing data values
- Structuring the dataset for machine learning models

These steps help improve the reliability and accuracy of the classification process.

Water Quality Classification Using Machine Learning

In this stage, **machine learning algorithms** will be applied to analyze the processed sensor data and classify water conditions. Models such as **Random Forest, Decision Tree, and Support Vector Machines (SVM)** will be used to identify patterns within the dataset.

The trained models will classify pool water conditions into three categories:

- **Safe** – Water parameters are within acceptable limits
- **Warning** – Some parameters show abnormal trends
- **Critical** – Water conditions may pose health risks

The classification results help pool operators quickly identify unsafe conditions and take corrective maintenance actions.

Water Quality Parameter Importance Analysis

After training the machine learning model, **feature importance analysis** will be performed to determine the relative contribution of each water quality parameter to the classification decision.

This analysis helps identify which parameters most strongly influence water safety, such as:

- Chlorine concentration
- pH level

- Turbidity
- Temperature
- Total dissolved solids (TDS)

Understanding parameter importance enables pool operators to focus on the most critical factors affecting water quality.

System Integration and Monitoring Interface

The final stage involves integrating the monitoring and analysis components into a **user-friendly monitoring system**. The system will present classification results and parameter analysis through a simple interface or dashboard.

Pool operators will be able to view:

- Real-time sensor readings
- Water quality classification results
- Important parameters affecting water safety
- Historical water quality trends

This interface allows operators to make informed decisions regarding chemical treatment, cleaning schedules, and overall pool maintenance.

2.3 System Architecture

2.3.1 Software Solution

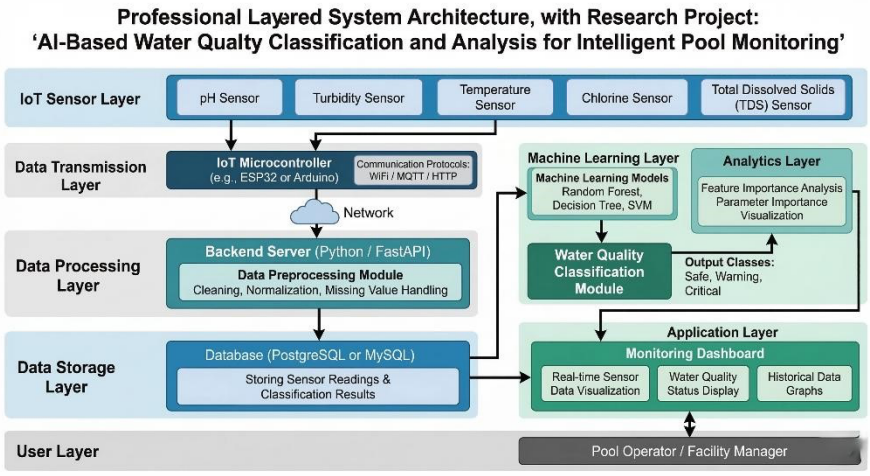


Figure 2: System Architecture Diagram

3.2 The Flow of the Project

3.2.1 Requirement Gathering and Analysis

The requirement gathering and analysis phase is a critical step in developing the AI-based water quality monitoring pool maintenance system. This stage focuses on identifying the requirements and expectations of stakeholders such as pool operators, facility managers, and system developers. Understanding these requirements helps ensure that the proposed system addresses real-world challenges related to pool safety, water quality monitoring, and efficient maintenance management.

Review of Literature and Existing Solutions

A comprehensive review of existing pool monitoring systems, IoT-based water quality monitoring technologies, and machine learning applications in environmental monitoring will be conducted. This review helps identify the strengths and limitations of current systems and highlights research gaps that the proposed solution aims to address. Both academic research papers and commercial monitoring solutions will be analyzed to understand the current technological landscape.

Stakeholder Feedback and System Requirements

To better understand practical challenges in pool maintenance, feedback from pool operators and facility management personnel will be considered. This helps identify issues such as delayed detection of water contamination, inefficient chemical usage, and limitations of manual testing methods. Stakeholders may

also provide insights regarding desired system features such as real-time monitoring dashboards and automated water quality classification.

Requirement Validation

The information gathered from literature review and stakeholder analysis will be organized into a detailed system requirement specification. Key system features such as real-time data monitoring, machine learning-based classification, and parameter importance analysis will be defined and prioritized. This step ensures that the system design directly addresses the most important operational and safety requirements of swimming pool management.

3.2.2 Feasibility Study

The feasibility study evaluates whether the proposed **AI-based water quality classification system** can be realistically developed and deployed. This evaluation considers **technical, operational, and time related feasibility factors**.

Schedule Feasibility

A detailed project schedule will be developed using project planning tools such as **Gantt charts**. The timeline will define key project stages including data collection, system design, machine learning model development, system integration, testing, and final implementation. Regular progress reviews will be conducted to monitor project milestones and ensure timely completion of each stage.

Technical Feasibility

The technical feasibility analysis examines whether the required technologies and tools are available for implementing the proposed system. The project will utilize **IoT sensors for data collection and machine learning frameworks such as Python-based libraries (Scikit-learn, TensorFlow, or similar tools)** for developing the classification models. Since these technologies are widely available and supported by existing research, the development of the proposed system is considered technically feasible.

Operational Feasibility

Operational feasibility focuses on whether the proposed system can be effectively used by pool operators and facility managers. The system will be designed with a **simple and user-friendly monitoring interface** that allows users to easily view water quality conditions and system recommendations. This ensures that the system can be practically integrated into routine pool maintenance operations.

- **Technical Feasibility:**
  The system requires advanced technical resources, including expertise in IoT sensor integration, machine learning model development, real-time data processing, and dashboard implementation. Tools such as Python, machine learning libraries, cloud platforms, and IoT frameworks will be

evaluated for suitability. The technical feasibility study will confirm that the project team has the required skills and infrastructure to build the system effectively.

- **Economic Feasibility:**

  A cost analysis will be conducted to ensure the project is financially sustainable. The analysis will estimate expenses related to IoT devices, cloud hosting, development tools, and human resources. To maintain viability, critical features such as real-time monitoring and predictive modeling will be prioritized, while cost-efficient approaches will be adopted for less essential components. This ensures that the system is both affordable and scalable without compromising quality..

3.2.3 Implementation

The implementation phase focuses on developing the **AI-based water quality classification module** for the intelligent pool monitoring system. During this stage, the proposed system architecture is transformed into a functional solution by developing and integrating key components such as sensor data collection, machine learning analysis, and monitoring interfaces. The goal of this phase is to ensure that the system operates reliably and provides accurate water quality analysis.

IoT Sensor Deployment and Data Collection

IoT-based sensors will be installed to continuously monitor important water quality parameters in the swimming pool. These sensors measure **pH level, turbidity, chlorine concentration, temperature, and total dissolved solids (TDS)**. The collected data will be transmitted to the processing system in real time.

The sensor data will be stored in a structured format so that it can be easily processed and analyzed. Continuous data collection allows the system to monitor changing water conditions and identify abnormal patterns.

Data Preprocessing and Feature Extraction

The collected sensor data will undergo preprocessing before being used in machine learning models. This step ensures that the dataset is accurate and suitable for analysis. The preprocessing process includes:

- Handling missing sensor values
- Removing noise and inconsistent readings
- Normalizing data values
- Structuring the dataset for model training

Relevant features from the dataset will then be extracted to improve the performance and accuracy of the machine learning models.

Water Quality Classification Modeling

Machine learning algorithms will be applied to classify the water quality conditions based on the processed sensor data. Models such as **Random Forest, Decision Tree, and Support Vector Machines (SVM)** will be trained using historical and real-time water quality data.

The trained models will classify water conditions into three categories:

- **Safe** – Water parameters remain within acceptable limits
- **Warning** – Some parameters indicate abnormal trends
- **Critical** – Water quality may pose potential health risks

This classification enables pool operators to quickly identify unsafe conditions and perform necessary maintenance actions.

Parameter Importance Analysis

After the classification model is developed, feature importance analysis will be performed to determine the influence of different water quality parameters on the classification results. This analysis helps identify the most critical factors affecting water safety, such as **chlorine levels, pH balance, turbidity, temperature, and TDS**.

Understanding parameter importance helps pool operators prioritize maintenance actions and optimize chemical usage.

Monitoring Dashboard and System Integration

Finally, the system components will be integrated into a **user-friendly monitoring interface**. The dashboard will display real-time sensor readings, water quality classification results, and parameter analysis. Pool operators will be able to monitor pool conditions, view historical data trends, and take appropriate actions to maintain safe water quality.

3.2.4 Testing

The testing phase is an important part of the development process to ensure that the **AI-based water quality classification system** operates accurately, reliably, and efficiently. Multiple levels of testing will be conducted to validate the functionality, performance, and usability of the proposed system.

Unit Testing

Unit testing will be performed on individual system modules to verify that each component functions correctly. This includes testing modules such as **sensor data collection, data preprocessing, machine learning model training, water quality classification, and parameter importance analysis**. Unit

testing helps identify errors at an early stage and ensures that each module performs its intended function before system integration.

Integration Testing

After individual modules have been tested, integration testing will be conducted to evaluate how different system components work together. This stage verifies that the workflow from **IoT sensor data collection to machine learning analysis and classification results** operates smoothly without system failures or data inconsistencies. Integration testing also ensures that data flows correctly between system components.

System Validation

To verify the effectiveness of the proposed system, validation will be performed by evaluating the performance of the machine learning models using **test datasets and evaluation metrics** such as accuracy, precision, recall, and F1-score. The classification results will be analyzed to determine whether the system correctly identifies **Safe, Warning, and Critical water conditions**.

Optimization and Refinement

Based on the testing results, necessary improvements will be made to enhance system performance. This may include refining the machine learning model, adjusting preprocessing techniques, and improving the monitoring interface. Continuous refinement ensures that the system provides reliable water quality analysis and supports efficient pool maintenance operations.

3.3 Project Requirements

3.3.1 Functional Requirements

The AI-based intelligent pool monitoring system must satisfy the following functional requirements to ensure effective water quality monitoring and analysis.

Real-Time Water Quality Monitoring

The system must continuously collect data from IoT sensors installed in the swimming pool environment. These sensors measure important parameters such as **pH level, turbidity, chlorine concentration, temperature, and total dissolved solids (TDS)**. Sensor readings should be accurate, reliable, and transmitted in real time to the processing module.

Water Quality Classification

The system must apply **machine learning algorithms** to analyze collected sensor data and classify pool water conditions into **Safe, Warning, and Critical categories**. The classification helps pool operators quickly detect abnormal conditions and take necessary maintenance actions.

Parameter Importance Analysis

The system must analyze the relative importance of different water quality parameters in the classification process. This feature identifies the most influential factors affecting water quality, such as chlorine levels or pH imbalance, enabling operators to focus on critical maintenance tasks.

Monitoring Dashboard

The system must provide a **user-friendly monitoring interface** that allows pool operators to view realtime sensor readings, classification results, and important water quality parameters. The dashboard should include visualizations, graphs, and historical data trends to support better decision-making.

Data Storage and Reporting

All water quality measurements, classification results, and system alerts must be stored in a secure database. The system should provide reporting tools that allow operators to analyze historical trends, track water quality changes, and evaluate maintenance performance.

System Integration

The system must integrate sensor data collection, machine learning analysis, and monitoring dashboards into a unified platform. The integration ensures that pool operators can monitor water quality and access system insights from a single interface.

3.3.2 Non-Functional Requirements

Reliability

The system must provide reliable monitoring and classification results to ensure accurate detection of unsafe water conditions.

Performance

The system should process sensor data efficiently and provide classification results in near real-time to support timely decision-making.

Usability

The monitoring interface must be simple and easy to use so that pool operators can quickly understand water quality conditions without technical expertise.

Scalability

The system should be designed so that additional sensors or monitoring locations can be integrated in the future without major system modifications.

Security

Water quality data and system information must be stored securely to prevent unauthorized access or data loss.

3.3.3 Software Requirements

To implement the **AI-based water quality classification module** for the intelligent pool monitoring system, the following software tools and frameworks will be required.

Python / Pandas / NumPy / Scikit-learn

Python will serve as the primary programming language for system development and machine learning model implementation. Libraries such as **Pandas and NumPy** will be used for data preprocessing and analysis of sensor readings. **Scikit-learn** will be used to train and evaluate machine learning models for water quality classification.

Machine Learning Frameworks

Machine learning libraries will be used to develop classification models such as **Random Forest, Decision Tree, and Support Vector Machines (SVM)**. These models will analyze sensor data and classify water conditions into **Safe, Warning, and Critical categories**.

Data Visualization Tools

Visualization libraries such as **Matplotlib and Seaborn** will be used to analyze water quality trends and display graphical insights during model evaluation and parameter importance analysis.

Database Management System

A relational database such as **MySQL or PostgreSQL** will be used to store water quality sensor readings, processed datasets, classification results, and historical records. This ensures reliable data storage and efficient data retrieval for analysis.

IoT Data Integration

IoT platforms or communication protocols will be used to transmit sensor data from monitoring devices to the processing system. This enables **real-time collection of water quality parameters** from the pool environment.

Development Environment

Tools such as **Jupyter Notebook, Visual Studio Code, or Google Colab** will be used for machine learning model development, experimentation, and testing.

Version Control System

Git and GitHub will be used to manage the project source code, track development changes, and maintain version control throughout the research project.

3. DESCRIPTION OF PERSONAL AND FACILITIES

Our project team consists of members with knowledge in **Artificial Intelligence, IoT systems, and software development**. Each member contributes specialized skills to different components of the intelligent pool monitoring system. Some team members focus on **sensor data collection and system integration**, while others are responsible for **machine learning model development, data analysis, and system implementation**.

For this research, modern computing resources will be used to develop and test machine learning models for water quality classification. Development tools such as **Python, machine learning libraries, and data analysis platforms** will support data preprocessing, model training, and evaluation.

In addition, cloud-based platforms and version control systems such as **GitHub** will be used for collaboration and project management among team members. Experimental environments will be created using **simulated water quality datasets and IoT sensor data**, allowing the team to evaluate the performance and accuracy of the classification models before real-world deployment.

These facilities and technical resources will ensure efficient system development, accurate analysis of water quality parameters, and reliable testing of the proposed intelligent monitoring solution.

4. COMMERCIALIZATION

The proposed **AI-based intelligent pool monitoring system** has strong potential for commercial application in sectors that require continuous monitoring of water quality. Target users include **swimming pools in hotels, sports complexes, recreational centers, schools, and public swimming facilities**.

The system provides several advantages for pool operators and facility managers:

- It enables **real-time monitoring of pool water quality** using IoT sensor technology.
- It applies **machine learning models to classify water conditions** into Safe, Warning, and Critical levels.
- It provides insights about **important water quality parameters**, helping operators take timely corrective actions.
- It reduces reliance on **manual water testing methods**, improving operational efficiency.

From an economic perspective, the system helps reduce maintenance costs by **detecting water quality problems early and optimizing chemical usage**. This leads to improved pool safety, better resource management, and enhanced operational efficiency.

Another key advantage of the system is its **scalability and adaptability**. The platform can be expanded to support additional monitoring features such as **automated alerts, mobile monitoring dashboards, and integration with smart facility management systems**.

With increasing demand for **smart infrastructure and automated monitoring technologies**, the proposed solution has strong potential for commercialization in the **hospitality, sports, and recreational industries**.

BUDGET AND BUDGET JUSTIFICATION

Component	Amount (LKR)
Cloud Hosting and Data Storage	9,000
Development Tools and Software Licenses	16,000

Machine Learning Model Trainning (Resources)	15,000
IoT Devices (Sensors & Connectivity)	25,000
Testing and Deployment	4,000
Maintenance and Support	5,000
Paper Publishing Cost	5,000
TOTAL	**79,000**

Table 2:Budget

5. REFERENCE LIST

[1] A. K. Singh, et al., "Water quality monitoring system using IoT and machine learning," *IEEE Access*, vol. 11, pp. 24560–24572, 2023. - https://ieeexplore.ieee.org/document/10698922

[2] M. S. Hossain, et al., "IoT-based smart water quality monitoring system for environmental management," *Sensors*, vol. 23, no. 4, pp. 1–18, 2023. - https://www.mdpi.com/2073-431X/13/10/267

[3] R. Kumar and P. Sharma, "Machine learning approaches for water quality prediction and classification," *Journal of Environmental Informatics*, vol. 40, no. 2, pp. 150–162, 2024.

[4] Y. Li, et al., "Smart water monitoring system using IoT sensors and cloud computing," *IEEE Internet of Things Journal*, vol. 10, no. 6, pp. 5231–5242, 2023. - https://ieeexplore.ieee.org/document/9936166

[5] S. Ahmed, et al., "Artificial intelligence techniques for water quality assessment: A review," *Environmental Monitoring and Assessment*, vol. 195, no. 7, pp. 1–15, 2023. - https://link.springer.com/article/10.1186/s42522-025-00170-8

[6] P. Mishra and R. Patel, "Real-time water quality monitoring using IoT and machine learning models,"

International Journal of Environmental Research, vol. 18, no. 2, pp. 255–266, 2024. - https://www.mdpi.com/1424-8220/24/9/2871

[7] J. Zhang, et al., "Machine learning-based water quality classification using sensor data," *IEEE Sensors Journal*, vol. 24, no. 1, pp. 1123–1132, 2024. - https://ieeexplore.ieee.org/document/10147310

[8] T. Nguyen and H. Tran, "IoT-enabled smart environmental monitoring systems: A comprehensive review," *Future Internet*, vol. 15, no. 3, pp. 1–20, 2023. - https://www.mdpi.com/2075-5309/16/3/656

[9] D. Wang, et al., "Feature importance analysis in machine learning for environmental monitoring systems," *Applied Artificial Intelligence*, vol. 38, no. 2, pp. 210–225, 2024. - https://www.tandfonline.com/doi/full/10.1080/00032719.2026.2618976

[10] K. Patel and S. Mehta, "AI-driven water quality prediction using sensor data analytics," *Journal of Environmental Engineering*, vol. 149, no. 6, pp. 04023021, 2023.

Gantt chart

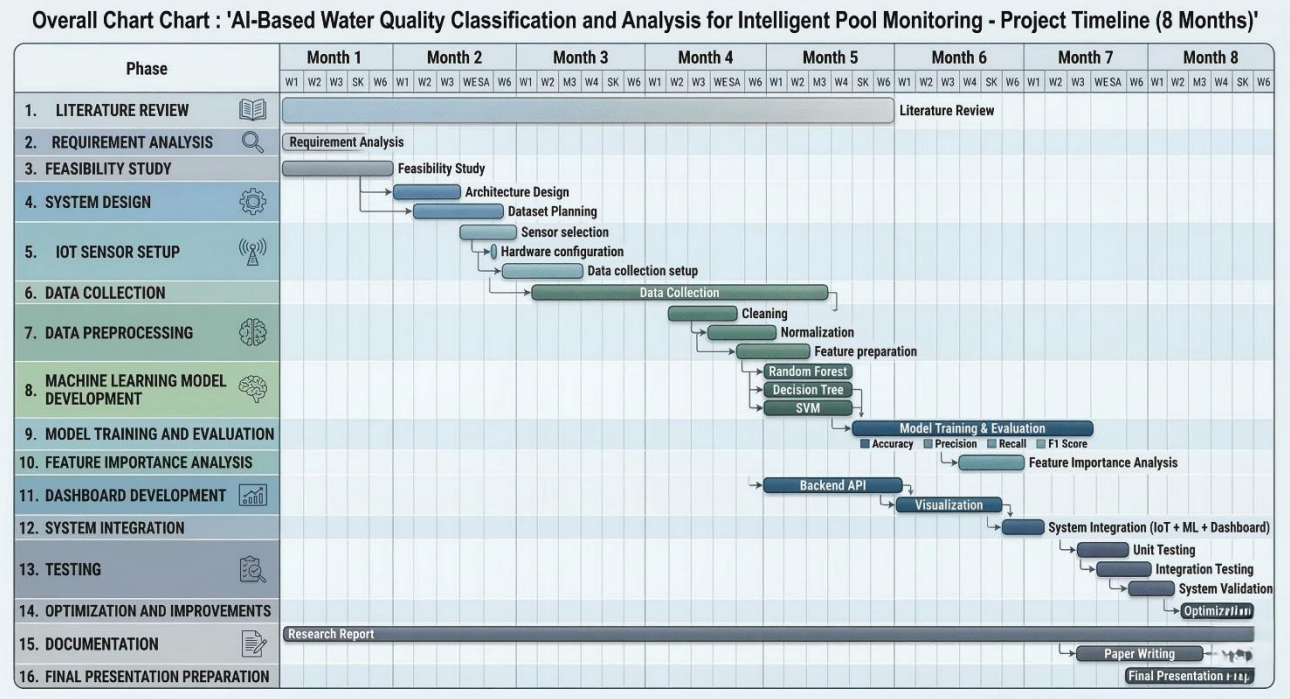


Figure 3 : Gantt Chart

6.APPENDICES

Appendix A – Water Quality Parameters Used for Classification

Appendix B – Dataset Used for Water Quality Prediction

Appendix C – Machine Learning Algorithms for Water Quality Classification

Appendix D – Water Quality Classification Process

Appendix E – Model Evaluation Metrics

Appendix F – Sample Water Quality Prediction Output

Work Breakdown Structure

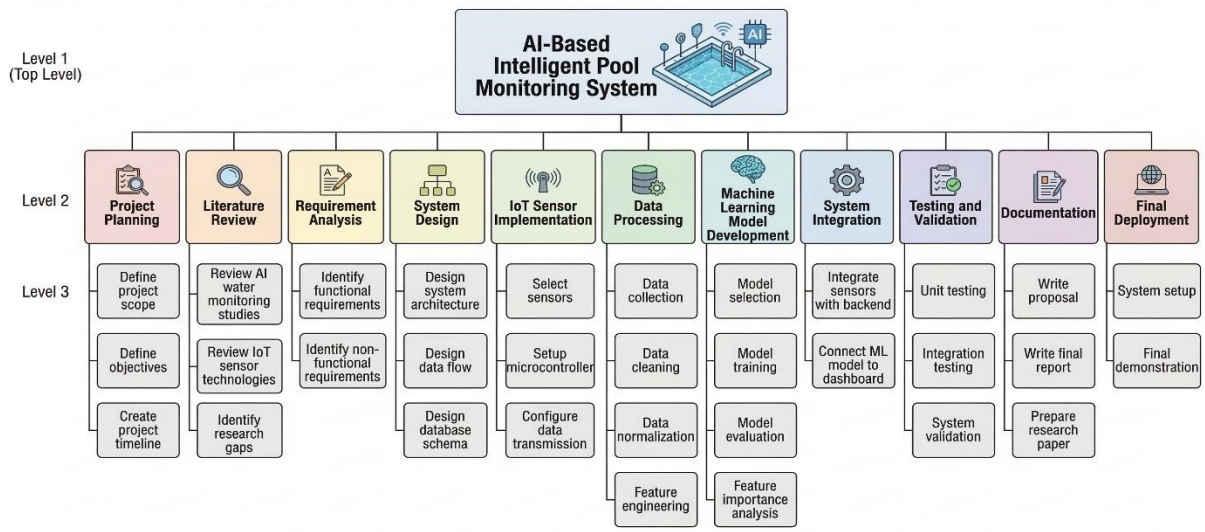


Figure 4: Work Breakdown Structure